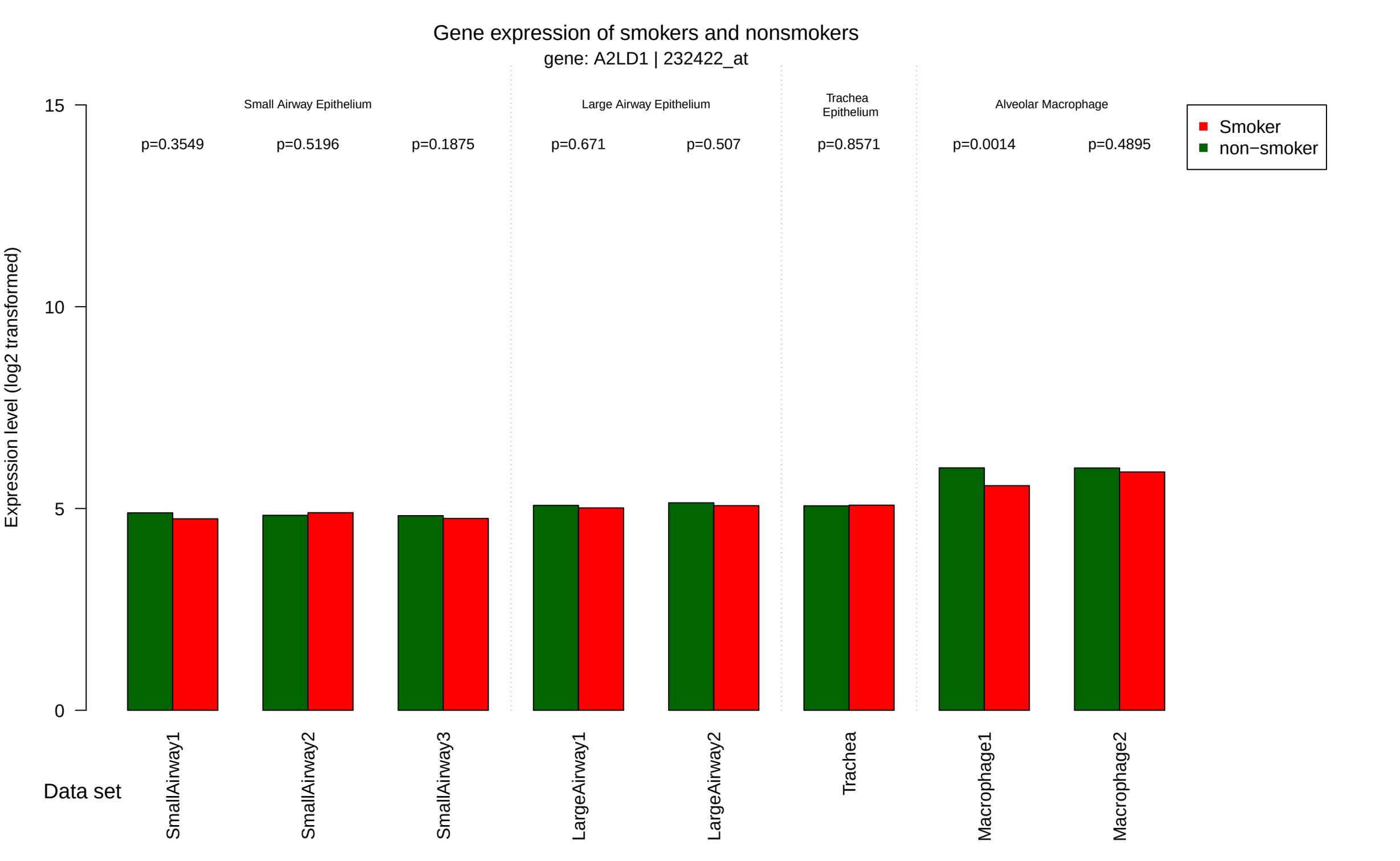
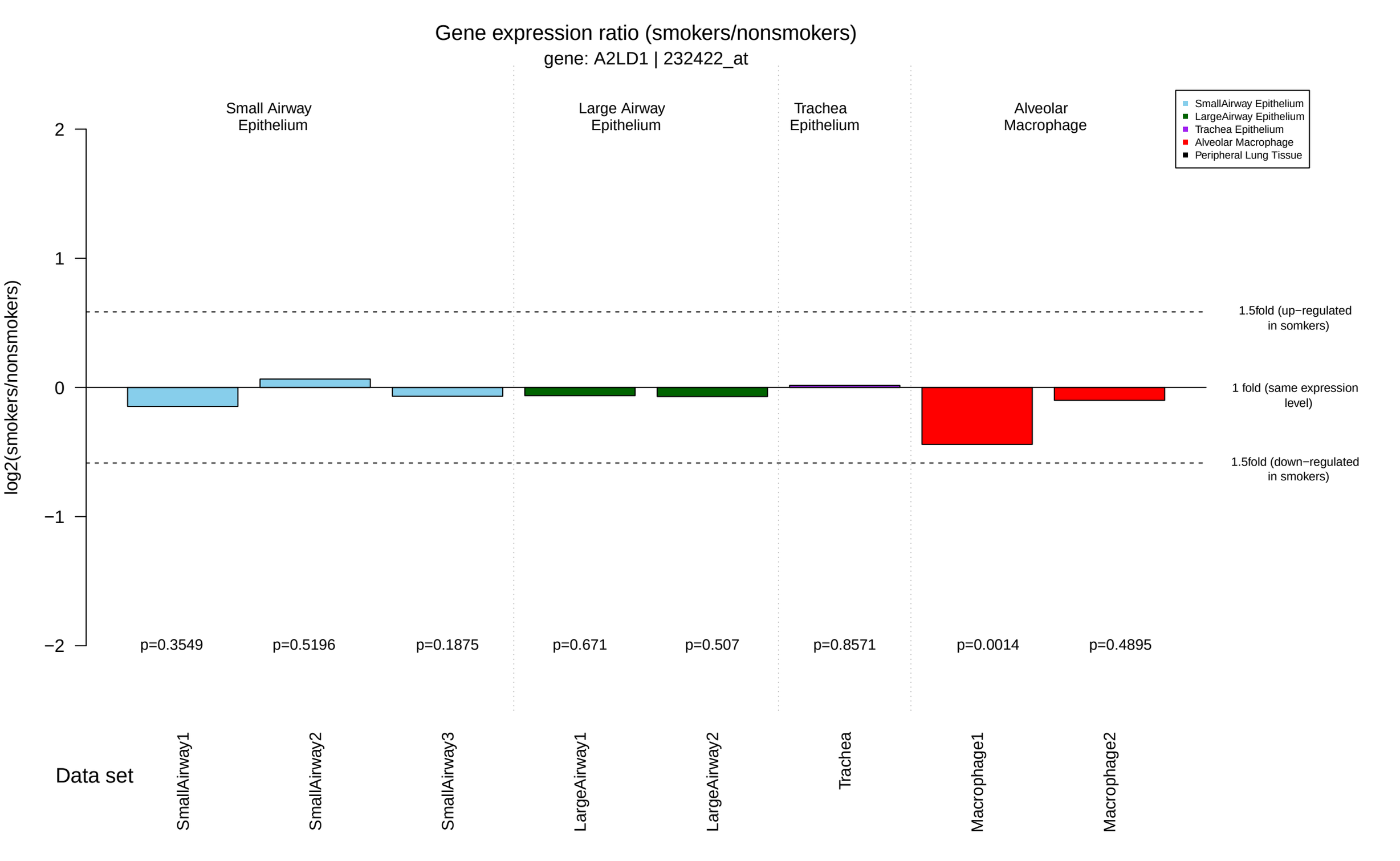
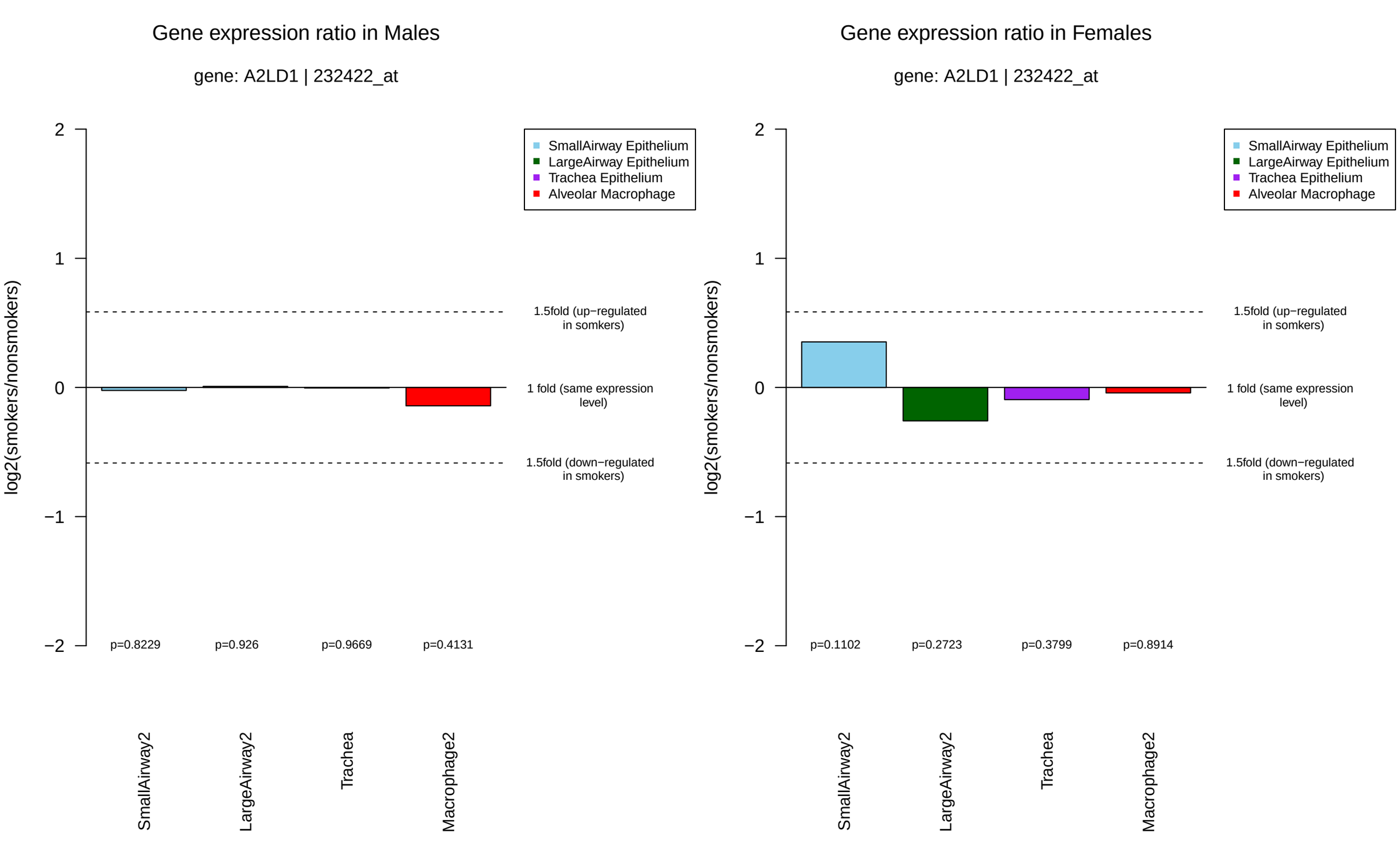
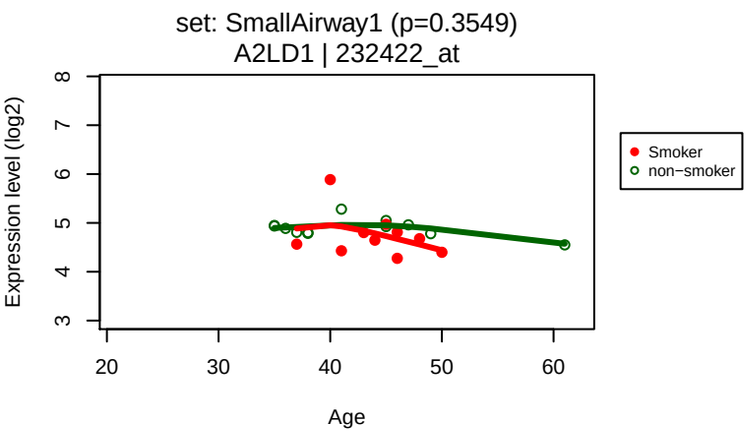
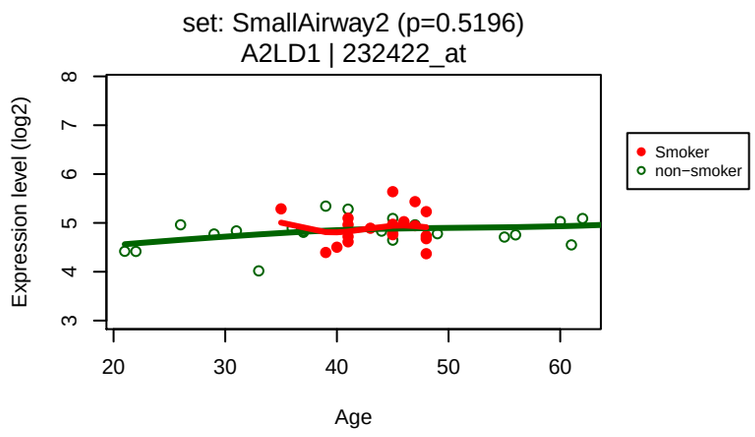
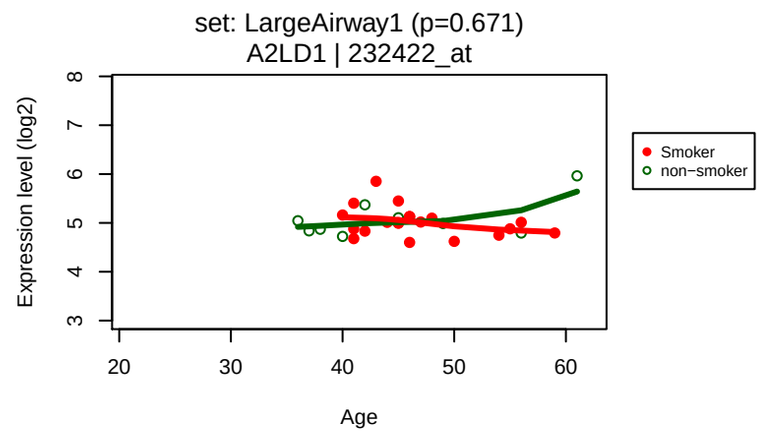

Gene expression of smokers and nonsmokers
gene: A2LD1 | 232422_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3549
p=0.5196
p=0.1875
p=0.671
p=0.507
p=0.8571
p=0.0014
p=0.4895
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

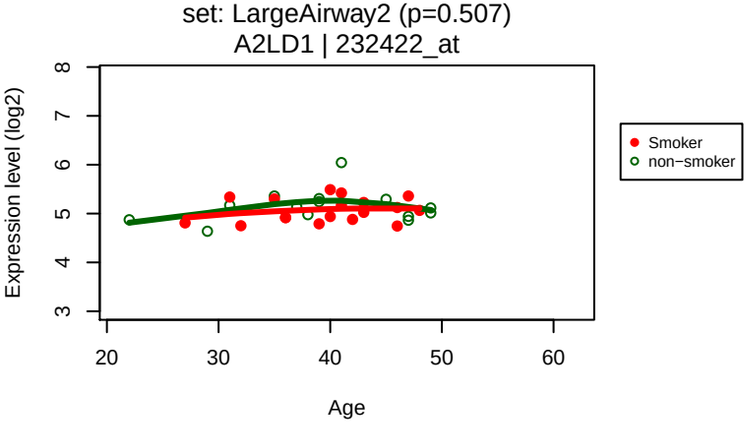

Gene expression ratio (smokers/nonsmokers)
gene: A2LD1 | 232422_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3549
p=0.5196
p=0.1875
p=0.671
p=0.507
p=0.8571
p=0.0014
p=0.4895
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

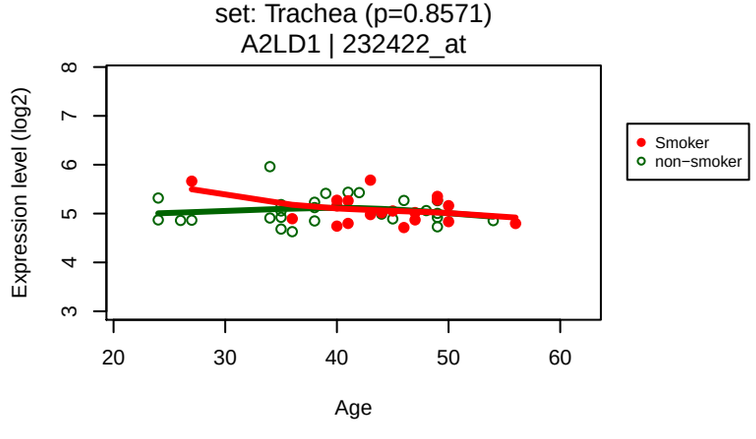

Gene expression ratio in Males
gene: A2LD1 | 232422_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8229
p=0.926
p=0.9669
p=0.4131
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: A2LD1 | 232422_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1102
p=0.2723
p=0.3799
p=0.8914
SmallAirway2
LargeAirway2
Trachea
Macrophage2

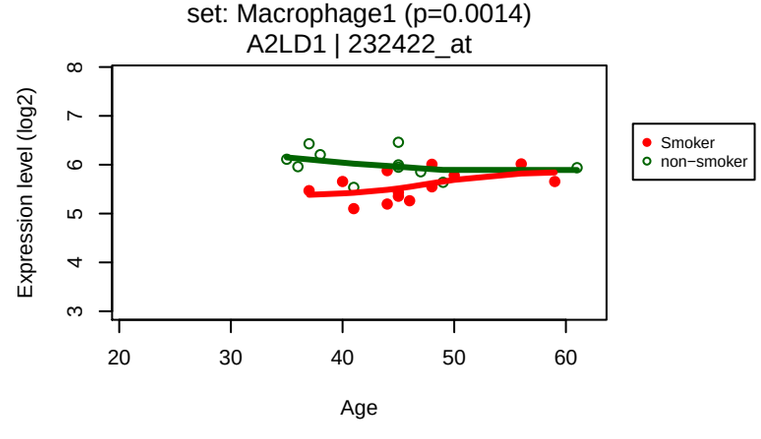

set: SmallAirway1 (p=0.3549)
A2LD1 | 232422_at
Expression level (log2)
Age
Smoker
non−smoker

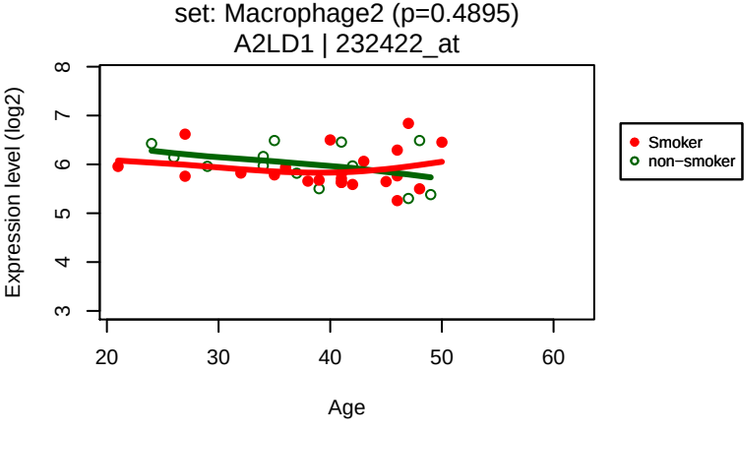

set: SmallAirway2 (p=0.5196)
A2LD1 | 232422_at
Expression level (log2)
Age
Smoker
non−smoker

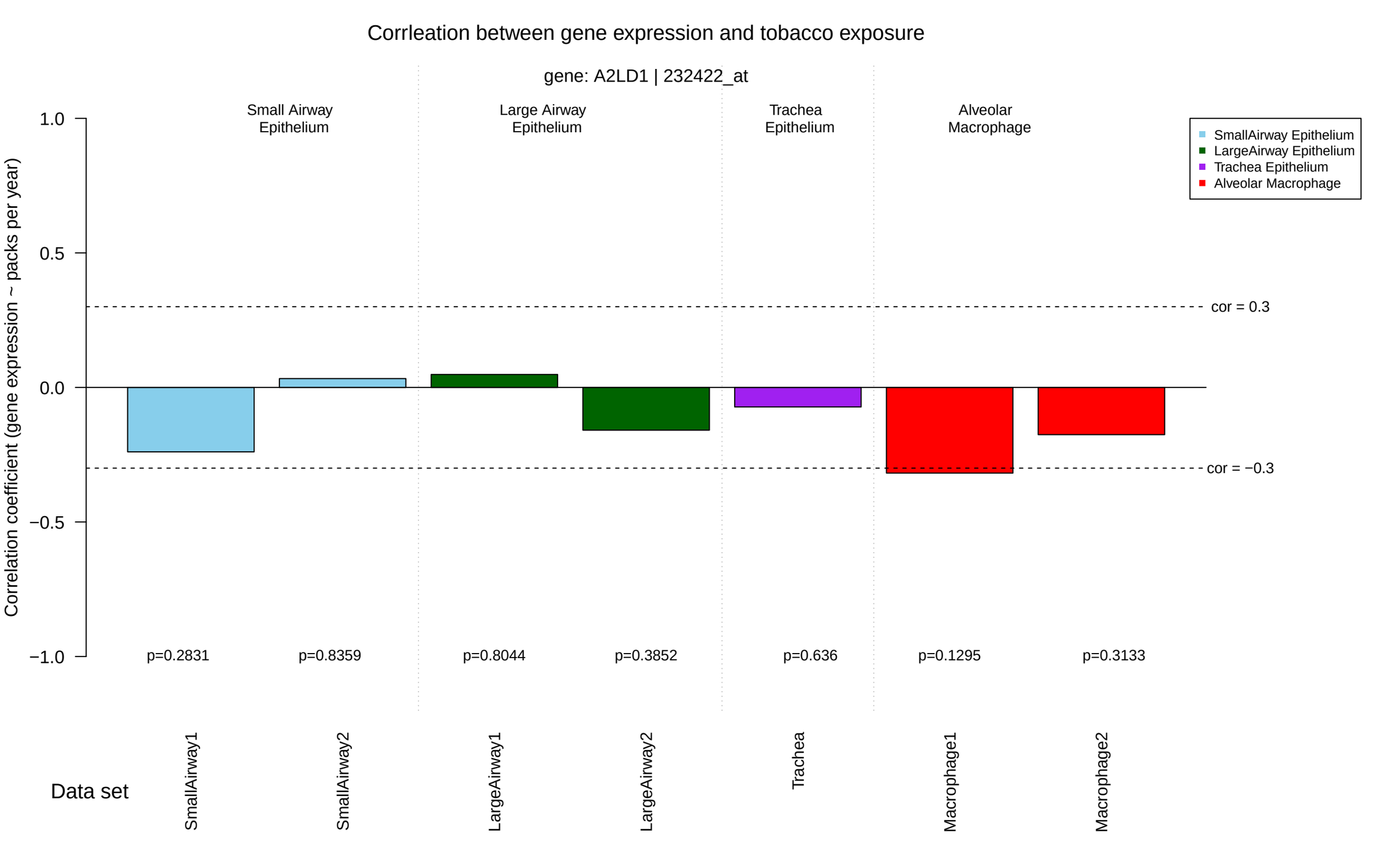

set: LargeAirway1 (p=0.671)
A2LD1 | 232422_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.507)
A2LD1 | 232422_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8571)
A2LD1 | 232422_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0014)
A2LD1 | 232422_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4895)
A2LD1 | 232422_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: A2LD1 | 232422_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2831
p=0.8359
p=0.8044
p=0.3852
p=0.636
p=0.1295
p=0.3133
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2